EXPERIMENT 20

Information Retrieval Using TF-IDF

Aim

To rank documents according to their relevance to a query using TF-IDF.

Procedure

1. Prepare a small collection of documents.
2. Create a TF-IDF vectorizer.
3. Transform the documents into vectors.
4. Transform the search query into a vector.
5. Calculate cosine similarity and rank the documents.

Program

```
# Install first: pip install scikit-learn
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.metrics.pairwise import cosine_similarity

documents = [
    "Python is useful for data science",
    "Football is a popular sport",
    "Python supports machine learning"
]
query = ["Python machine learning"]

vectorizer = TfidfVectorizer()
matrix = vectorizer.fit_transform(documents + query)
scores = cosine_similarity(matrix[-1], matrix[:-1]).flatten()

for index in scores.argsort()[::-1]:
    print(f"Document {index + 1}: score={scores[index]:.3f}")
```

Output

```
OUTPUT — EXPERIMENT 20
Document 3: score=0.5...
Document 1: score=0.2...
Document 2: score=0.0
(Exact scores depend on the TF-IDF implementation.)
```

Result

Thus, documents were ranked by their TF-IDF vector similarity to the query.